

Class 18: Defining Determinants

The determinant is a function that takes a square matrix A as its input and produces a number as its output. The value of the determinant is defined iteratively in terms of smaller submatrices.

Notation and Terminology

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 1 & 2 \\ 4 & 3 & 1 & 1 \end{bmatrix}$$

1. A_{ij} is obtained by

$$A_{31} =$$

2. $\det A_{ij}$ is called

3. $(-1)^{i+j} \det A_{ij}$ is called

Definition: If $A = [a]$ then $\det(A) =$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $\det A =$

If A is $n \times n$ then $\det A =$

Notation: $\det A$ is often denoted by $|A|$ for compactness.

Example: Find a formula for the determinant of a 3 x 3 matrix by expanding along the first row. Choose any other row or column to expand along and find $|A|$.

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} =$$

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} =$$

Theorem: The cofactor expansion for the determinant is a matrix produces the same value using any row or any column.

Strategy when using cofactor to compute determinants

Example: $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & 3 \end{vmatrix}$

Practice:

a. $\begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix}$

b. $\begin{vmatrix} 1 & 0 & 4 \\ 2 & 0 & 5 \\ 3 & 0 & 6 \end{vmatrix}$

c. $\begin{vmatrix} 1 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 \\ 3 & 6 & 8 & 0 \\ 4 & 7 & 9 & 10 \end{vmatrix}$

Questions:

How does the determinant of A^T compare to the determinant of A ?

How does the determinant of AB compare to the determinant of A and determinant of B ?

How does the determinant of A^{-1} compare to the determinant of A ?